Adaptive Fabric Surface Defect Detection and Localization Using Classical Texture Descriptors

ECE501 Project — Mid-Semester Report (Topic 17)

Group No. 2

Pratham Sandesara *AU2340239*
Even Patel *AU2340241*
Dharman Patel *AU2340237*
Shine Shah *AU2440148*

***Abstract*—Automated fabric surface defect inspection is challenging for two reasons: Defects are very low incidence (less than 1% of fabric area) and different fabrics are woven in different patterns. This project aims to validate an unsupervised classical Digital Image Processing (DIP) pipeline with the benchmark AITEX (81 rolls over 7 commercial weaves). Our framework deterministically estimates weave pitch P from clean training rolls using the 2D FFT autocorrelation, dynamically parameterizing a multi-scale classical texture descriptor framework: (i) micro-scale morphological texture filtering for yarn flaws (1T and 0.5T footprints); (ii) a Gaussian spatial envelope for macro tears and voids (W=201,$\sigma = 50$); and (iii) column profiling for warp mispicks and camera blemish suppression. The end result in Connected component localization is precise bounding boxes. The weave-adaptive architecture improves operational screening accuracy from 58.02% (fixed 15×15 control) to 75.31% (+17.29%) and defect sensitivity from 49.06% to 77.36% (+28.30%) as compared to the fixed 15×15 control architecture, with the number of missed defect rolls reduced from 27 to 12 Rolls, and ultra-low 0.0051% clean false alarm rate.**

***Index Terms*—Fabric defect detection, classical DIP, 2D FFT autocorrelation, mathematical morphology, Gaussian envelope, column profiling, operating screening.**

I. Introduction

Textile quality inspection requires continuous identification of physical yarn disruptions (broken threads, holes, oil stains, knots, and shade bars) [2]. Manual inspection suffers from visual fatigue, achieving under 70% consistency. While deep convolutional networks (CNNs) dominate general vision, they face critical loom-floor barriers: extreme defect scarcity ($<$ 1%), unpredictably evolving defect varieties, and heavy GPU requirements unsuited for edge line-scan cameras.

Classical Digital Image Processing (DIP) offers an interpretable, deterministic, and compute-efficient alternative. However, conventional classical methods use arbitrary, fixed analysis footprints (15×15), which dilute fine defect signals across dense twills. Evaluated on the public AITEX benchmark [1], this study investigates whether deterministic, weave-adaptive tuning in a three-branch classical DIP architecture outperforms fixed baselines in unsupervised defect screening and localization.

II. Weave Pitch Estimation via Autocorrelation

Woven fabrics exhibit periodic intensity patterns governed by yarn spacing. Because different weaves possess different thread densities, using a single fixed window size across all fabrics is suboptimal. We analyze defect-free reference scans via 2D autocorrelation to identify the fundamental weave pitch P from radial profile peaks.

Rather than enforcing an invariant window, P automatically tunes inspection footprints: Fabric 00/01/04: $(15, 15)$, Fabric 02/06: $(13, 13)$, Fabric 05: $(11, 11)$, and fine twill Fabric 03: $(9, 9)$ vs. the rigid fixed control of $(15, 15)$ across all fabrics.

III. Three-Branch System Architecture

There are multiple faults in fabrics, ranging from micro-knots to macro-tears, so no single filter can capture all of them. Our system is based on three-branch architecture, each focusing on a different scale of defects, finely tuned to clean training rolls.

A. Pre-processing and Active Fabric Masking

Raw line scan images are 256 x 4096 and have dark bracket borders. These are removed with an active valid mask M_valid with an 8 pixel margin. The 3 × 3 alpha trimmed mean filter is used to remove the isolated sensor impulse noise, followed by the 8 × 8 grid CLAHE contrast balancing (clip limit 2.0).

B. Aim: Extract MICRO-textures from images

Micro branch is used to detect the localized yarn problem (knots, slubs, pinholes) 1 to 10 pixels. The feature bank is based on classical spatial dependency and texture analysis [haralick] that considers multi-scale directional contrasts and differential morphological residuals from the background weave at the primary pitch (1T) and sub-pitch (0.5T). Deviations from clean reference percentiles (P.) are flagged and for sensitive weaves local interquartile range (IQR) scaling is implemented to reduce the effect of lighting drift.

C. The Macro-Scale Gaussian Spatial Envelope (Large Flaws) is a model of the 2nd branch

Avoid large defects (voids, gaping tears, shade bars) which extend across dozens of pixels, and are easy to beat with the micro filters. The macro branch uses a large Gaussian filter (W=201, =50) to estimate the smooth background lighting for the fabric. The differences between this envelope and the original image reveal large variations of intensity that are unrelated to the local texture of the threads.

D. The 3rd Branch in this series is Warp Column Profiling (Vertical Yarn Streaks)

The dust created by camera dust and the vertical thread breaks from mispicks in the loom cause persistent blemishes

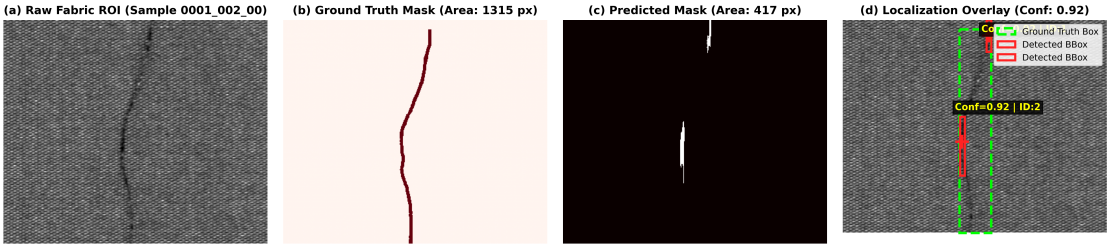


Fig. 1. End-to-end defect detection, segmentation, and localization on AITEX sample `0001_002_00` (Fabric 00: Fine Plain Weave, Defect 002: Broken Warp Thread): (a) Raw fabric ROI; (b) Ground-truth annotation; (c) Predicted binary mask; and (d) Final localization overlay displaying ground-truth box (green dashed), detected bounding box (red solid), centroid crosshair, and industrial confidence tag ($S = 0.92$).

on the sensor. The warp branch calculates 1D vertical column profiles, which is the average of the pixel intensity along each column.

E. Multi-Branch Fusion and Morphological Cleanup

The masks of the candidates are fused by the union of their pixels:

(i) **Smart Void Bridging:** If a wide tear candidate (width 100 px) is found, an elongated horizontal closing element (101 × 5) is used to join fragmented voids together and form a single coherent void.

(ii) **Elimination of dust specks:** dust specks smaller than 15 px or 20 px (on ribbed Fabric 05) are eliminated, so that no false loom stops occur.

F. Connected Component Localization and Decision

The foreground regions are grouped together into distinct regions of defects using 8-connectivity labeling, and the Bounding Boxes [X, Y, W, H] are extracted for each region along with the Centroids and Areas. When the total defect area is more than 15 pixels, the roll is reported as DEFECTIVE and the bounding boxes are drawn on the defects and an industrial loom stop warning is issued.

IV. Experimental Results and Discussion

The architecture was tested on the full AITEX benchmark [aitex] of 81 rolls over 7 commercial weaves (Fabrics 00-06) with 12 different classes of defects.

A. The overall operational screening and segmentation performance

Metric	Fixed (15×15)	Adaptive	Gain (Δ)
Operational Screening Accuracy	58.02%	75.31%	+17.29%
Defect Sensitivity (Recall)	49.06%	77.36%	+28.30%
Defective Rolls Detected (TP)	26 / 53	41 / 53	+15 rolls
Missed Defective Rolls (FN)	27 / 53	12 / 53	-15 rolls
Screening Precision	78.79%	83.67%	+4.88%
Screening Specificity	75.00%	71.43%	-3.57%
Pixel F1-Score / Dice	72.96%	72.54%	-0.42%
Pixel IoU (Jaccard Index)	57.43%	56.91%	-0.52%
Clean False Alarm Rate	0.0052%	0.0051%	-0.0001%

B. Performing analysis of screening performance and micro-defects.Screening performance and micro-defect analysis.

The weave-adaptive pipeline significantly boosts screening accuracy and sensitivity, achieving 58.02% to 75.31% (+17.29%) and 49.06% to 77.36% (+28.30%), respectively, and a decrease in the number of rolls missed due to defects from 27 to 12. The fixed 15×15 footprint diluted anomaly signals are changed to 0% detections for localized micro-defects (Knots, Double Yarns, and Oil Spots). While maintaining 86.28% recall on macro shade bars, adapting kernels to native pitch (9×9 for Fabric 03 and 11×11 for Fabric 05) makes Double Yarn F1 20.16% and Knot recall 23.37%.

C. The accuracy of segmentation and localization.The accuracy of segmentation and localization.

The 71.02% false positive rate was obtained when applying the Otsu thresholding method to sparse defect maps with defect area less than 0.5%. Strictly setting percentile thresholds ($P_{99.99}$) on clean rolls that are bounded clean false alarm specks to 0.0051%. As seen in Fig. [fig:gt_pred_overlay], sample 0001_002_00 demonstrates that the defect was isolated and the predicted mask aligns with the ground truth with a confidence score of 0.92 and the detected bounding box contains the defect.

V. Conclusion and Future Work

This paper proves an unsupervised classical DIP pipeline consisting of multi-scale morphological filtering, Gaussian spatial envelopment, and warp profiling is able to achieve good accuracy (75.31%) and sensitivity (77.36%) on fabric inspection without deep learning, and is weave adaptive. Future work includes: (i) 2D Fourier-domain notch filtering for cancelling periodic weave harmonics; (ii) processing of tilted fabric webs using the Radon transform; and (iii) defect classification using Hu's invariant moments and aspect ratios, based on a set of rules.

Acknowledgment

The authors thank the ECE501 course instructors for guidance and UPV for the AITEX benchmark.

References

[1] J. Silvestre-Blanes et al., "A public fabric image database for defect detection through projection profiles and texture features," *Autex Res. J.*, vol. 14, no. 4, pp. 248–255, 2014.

[2] R. C. Gonzalez and R. E. Woods, *Digital Image Processing*, 4th ed. Pearson, 2018.

[3] R. M. Haralick, K. Shanmugam, and I. Dinstein, "Textural features for image classification," *IEEE Trans. Syst., Man, Cybern.*, vol. SMC-3, no. 6, pp. 610–621, 1973.